

THE BLOGGER GRANNY

Phyllis Greene can hardly take care of herself but at the age of 90, she decided to start a blog. She talks about her near addiction and how technology has brought a new dimension to her life. Worth a read to try and understand the elderly.
<http://bbc.in/ddebD8>

SPEED TEST: PIGEON V/S BROADBAND

A showdown between the juxtaposed modes of communication highlighted the pathetically low speeds of rural broadband in the UK.
<http://bbc.in/9wcTXu>

**KEYLESS KEYBOARDS**

SnapKeys believes that screens were not meant to take inputs. So instead of trying to port qwerty to the screen, they've come up with a system that involves has four invisible keys – two on each side of the device's screen. Worth a look.
<http://bit.ly/dfahfk>

GOOGLE EMPLOYEES CAN SNOOP ON YOUR EMAILS!

If you've ever had this nagging doubt while using the service, it's not paranoia. A story on The Independent reports how a Google engineer was accessing several accounts including that of a 15 yr old boy. Pervert!
<http://bit.ly/ca6g3p>

Sandy Bridge: Intel's Next-Generation Microarchitecture Revealed

by Matthew Murray



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At this week's Intel Developer Forum in San Francisco, Intel wasted no time revealing the most anticipated technology of the three-day conference: its second-generation Core microarchitecture, code-named Sandy Bridge. It's due for both desktop and laptop release at the first part of next year, but has been generating buzz for months as tantalizing little details have slowly leaked out. When it came to Sandy Bridge, we wanted to know two things: What exactly was it, and was it worth all the hype? The first question has been definitively answered—and we think we've got a good start on the second.

At its most basic, Sandy Bridge is the latest "tock" in Intel's tick-tock develop-

ment strategy: the latest iteration of the Westmere-based, 32nm die-shrink of the Nehalem microarchitecture that Intel released at the beginning of the year. They won't fit the LGA1156 socket those chips use, however—they'll need a new one, LGA1155, which will be released on motherboards using the new Series 6 chipset at the same time Sandy Bridge CPUs are released.

But beyond all that, Sandy Bridge is really about unity. Yes, it finally brings the memory controller, PCI Express (PCIe) controller, and video functions within the processor die, which by itself is an enormous change that has major performance repercussions. But it also strikes up a much clearer parity with the modern computing world: finding new ways, for example, to take advantage of large numbers of processing cores (neces-

sary, as Intel released the first consumer-oriented six-core processor earlier this year), and use less power to do everything.

Most users won't notice these changes, but many of them are substantial ones that could be setting the ground for even more exciting developments to come. Read on for our look at what's going on in Sandy Bridge, and what benefits it may hold for computer users and builders starting in 2011.

Front end features

Two of the major changes that Sandy Bridge introduces come right at the front end: a new decoded micro-operation (uop) cache and revamped branch prediction.

Micro-operations and AVX

Sandy Bridge now treats where you want to store data and what data you want to store as separate functions. (Intel began implementing this idea in Nehalem.) When an instruction is decoded, the uop cache checks to see whether it's already cached. If the instruction is, the cache shuts down the rest of the pipeline and the front end. If it's not, the instruction is cached; the cache can store up to 1,500 instructions, and is direct-mapped for easier access. If an instruction hasn't been used in a while, it's discarded once additional space is needed. Because the uop cache has marginally higher and more consistent bandwidth than the instruction cache, this one-two punch has a significant impact on both power and performance.

In addition, the use of a new physical register file instead of a centralized retire-

